

Which Part Comes from Distinction and Which from the Trace: A Response on *Arithmetic from Distinction*

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Abstract

A careful reader of *Arithmetic from Distinction: A Number Tower Forced by a Single Primitive* raises the right objection. The trace construction (ε , the extension rule $\tau \mapsto \tau \cdot \delta$, concatenation, length) is a free monoid on one generator, which is already a unary copy of the natural numbers; so it looks as if arithmetic follows from the trace monoid rather than from distinction, and the requirement in Section 3 that the sameness relation $\sim$ be an equivalence looks like a second, independent assumption. This note answers the objection directly. The short version: the reader has correctly identified the load-bearing claim, and that claim is a theorem, not a rhetorical move. The free monoid on one generator is the *initial* object in the category of pointed endomaps; initiality is exactly what “forced” means; and the trace structure is therefore not one formalization of distinction among many but the unique relation-free one, into which every other realization maps canonically. The equivalence-relation requirement on $\sim$ is likewise derived rather than posited: any tight, separating judgment is forced to coincide with the decidable equality the act already carries. Both facts are formalized in Lean 4 with no project-local axioms and no **sorry**. I give the precise statements, the module names, and the exact boundary where the forcing stops, which is the part of the question that I think makes the central claim defensible rather than overstated.

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1 The question, stated fairly

The objection has three parts, and I will keep them separate because they have different answers.

- (Q1) **The trace monoid does the work.** The paper opens with δ as an atomic act, then immediately introduces traces with the empty trace ε and the extension rule $\tau \mapsto \tau \cdot \delta$, together with concatenation, recursion, induction, and length. Traces generated by ε and $\tau \cdot \delta$ are a unary representation of $\mathbb{N}$. Once that successor structure is present, addition, multiplication, order, and induction are standard. So “arithmetic from distinction” looks closer to “arithmetic from a free monoid generated by one distinction symbol,” and the word *distinction* is doing rhetorical rather than mathematical work.
- (Q2) **Which part is forced?** Granting traces, extension, and orbit length, arithmetic follows naturally. So which part comes from distinction and which from the trace construction?
- (Q3) **The equivalence relation is an extra assumption.** Section 3 *requires* $\sim$ to be reflexive, symmetric, and transitive. Why should distinction necessarily produce exactly this trace structure, and exactly this relation, rather than some other formalization? The paper proves arithmetic follows from the orbit structure, but does not, within its own four corners, prove that the orbit structure is forced by distinction.

I accept the framing. In fact I will go further than the reader and concede the exact sub-claim that is correct, because the honest concession is what makes the real answer land.

2 What the objection gets right

Read strictly within the four corners of *Arithmetic from Distinction*, the reader is correct. Definition 2.2 posits two formation rules for traces, and the collection they generate, under concatenation, is the free monoid on one generator. The free monoid on one generator is $(\mathbb{N}, 0, +)$ up to unique isomorphism, and its element-counting map is the successor. The paper even says so out loud: it quantifies over traces “exactly as one quantifies over the natural numbers generated by zero and successor.” So a referee who asks “where did the successor come from?” and answers “from Definition 2.2, not from δ ” has read the paper accurately. That paper, taken alone, establishes *arithmetic from the free monoid on one generator*, and it does not, on its own pages, prove that “one generator” is forced by “distinction” against the alternatives. The same is true of Section 3: it *requires* $\sim$ to be an equivalence rather than deriving the requirement.

This is a real gap in the paper’s self-containment, and the reader is right to flag it. It is not a gap in the program. The missing steps are theorems, they are formalized, and the rest of this note states them. The fix to the paper is editorial: cite the theorems where the paper currently asserts. I treat that as part of the work still to complete (Section 9).

3 Freeness is forcing: the initiality theorem

The reader’s own phrase, “a free monoid generated by one distinction symbol,” is the answer, once one takes the word *free* seriously.

A structure is *free* on a generating datum when it imposes no relations beyond those the datum forces. Equivalently, in categorical language, it is the *initial* object in the appropriate category. The relevant category here is pointed endomaps: objects are triples (X, x, f) with a carrier X , a base point $x \in X$, and an endomorphism $f : X \rightarrow X$. A triple (N, z, s) is a *Lawvere natural-number object* (NNO) when for every (X, x, f) there is a unique $h : N \rightarrow X$ with $h(z) = x$ and $h \circ s = f \circ h$. This is the statement “ N is the initial pointed endomap,” and it is what “ $\mathbb{N}$ ” means in any foundation that does not presuppose $\mathbb{N}$.

Theorem 3.1 (Trace orbit is a natural-number object). *The trace orbit, with ε as base point and $\tau \mapsto \tau \cdot \delta$ as endomap, is a Lawvere natural-number object: primitive recursion into every pointed endomap exists and is unique.*

Formalized as `logicNat_isNNO` in `Foundation/UniversalForcing/NaturalNumberObject.lean`. The four NNO fields (`recursor`, `recursor_zero`, `recursor_step`, `recursor_unique`) are discharged by structural induction on the orbit.

Theorem 3.2 (Uniqueness up to canonical isomorphism). *Any two natural-number objects are canonically isomorphic: from the two universal properties one builds maps each way and proves both composites are the identity, again by the uniqueness clause.*

Formalized as `IsNaturalNumberObject.equiv`. Together with Theorem 3.1 this is the precise content of “forced.” The trace monoid is not a formalization of distinction chosen against rivals. It is the initial one. Initiality says there is a unique structure-preserving map *out* of it into every pointed endomap, so it sits canonically inside the landscape of all such structures rather than beside them. Any alternative formalization of “one act, iterated” is one of two things. If it is also free, it is isomorphic to the trace monoid by Theorem 3.2, so it is the same object. If it is not free, it is a proper quotient, which means it *imposes a relation*: it declares two distinct iteration counts to be the same. But declaring two distinct counts to be the same is an act of identification that “distinction” does not supply. It is extra information. So the relation-free realization, the one that adds nothing to “an act was performed and recorded,” is forced to be $\mathbb{N}$.

This is the direct answer to (Q3)’s last sentence. “Why this structure and not another?” Because every other structure on the same generating datum either is this one or requires identifications that distinction alone cannot license. The minimal, assumption-free realization is unique, and it is the number tower.

4 The successor is derived, not posited

Theorem 3.1 still defines the orbit syntactically, by the two trace rules, so a determined skeptic can repeat (Q1): you wrote the successor into the formation rules. The deeper layer of the formalization removes even that. It never mentions traces.

Take any realization with a carrier C , a native comparison cost `compare` : $C \times C \rightarrow \text{Cost}$, a native composition, a base element `one`, and a distinguished generator. Let `interpret`(n) be the n -fold composition of the generator starting at `one`. Assume one law, and only one:

$$(\text{recognition law}) \quad \text{compare}(\text{interpret}(m), \text{interpret}(n)) = 0 \implies m = n.$$

In words: distinct acts of distinction are distinguishable. The cost between two different iterates does not vanish.

Theorem 4.1 (Forced faithful $\mathbb{N}$). *The recognition law implies `interpret` is injective. Hence the carrier contains a faithful copy of $\mathbb{N}$ whose zero is `one` and whose successor is composition with the generator. The copy is derived from the law, not supplied as data.*

Formalized as `interpret_injective` (and the derived equivalence `nativeOrbitEquiv`) in `Foundation/Universal`. The proof is two lines: if `interpret`(m) = `interpret`(n) then by reflexivity of the cost (the always-present `identity_law`) the comparison is zero, so the recognition law gives $m = n$.

Here the successor is not a formation rule. It is the image of the generator under iteration, and the fact that it behaves like the Peano successor, in particular that it never wraps around or collapses, is *forced* by the recognition law. The honest motivation block at the head of that module records why this matters: the earlier strict layer derived its orbit “by construction”

($\mathbb{N} \simeq \mathbb{N}$), so it forced nothing from the operators; Theorem 4.1 replaces the by-construction copy with one the recognition law produces from the realization’s own comparison.

The sharpest version of (Q1) is answered by a single example inside the same file. Consider the discrete Boolean realization, whose carrier is $\{0,1\}$ and whose step is negation. The interpretation map is parity, $\text{interpret}(n) = \text{bodd}(n)$, formalized as `interpret_eq_parity`, and it is *not* injective (`interpret_collapses`). The carrier has two elements. Yet the iteration object is still the full natural-number object (`boolean_freeOrbit_isNNO`). This dissociates “the orbit structure” from “the syntax of traces” as cleanly as one could want. The orbit is whatever the universal property names, and it is $\mathbb{N}$ even when the carrier it acts on has been crushed to two points. Counting survived the collapse; the carrier did not.

5 It is the same $\mathbb{N}$ under the real cost, not a toy

One could still worry that “distinct distinctions are distinguishable” is a degenerate equality test, true only because the toy realizations use the equality cost if $a = b$ then 0 else 1. It is not. The same forcing runs under the actual recognition cost.

Let $J(x) = \frac{1}{2}(x+x^{-1})-1$, the cost the Law of Logic forces (uniqueness theorem `Cost.FunctionalEquation.law` and the completeness-free re-derivation in `PRCNativeCostUniqueness`). Build the multiplicative ladder on positive reals with doubling generator, compared by J of the ratio. Because $J(x) = 0 \iff x = 1$, distinct rungs $2^m, 2^n$ compare to zero only when $m = n$.

Theorem 5.1 (The recognition cost forces the arithmetic). *The J -cost ladder satisfies the recognition law, so its forced native $\mathbb{N}$ -copy is genuine for the canonical cost, not only for the degenerate equality cost.*

Formalized as `jcostLadder_faithful` in the `JCostLadder` namespace of `FaithfulOrbit.lean`. The number tower is the same object whether the comparison is a placeholder or the cost forced by the functional equation.

6 The equivalence relation is forced, not assumed

Now (Q3) on its own terms. Section 3 of the paper *requires* $\sim$ to be reflexive, symmetric, and transitive, and the referee reads this as a second primitive. In the formalization it is a theorem.

Model a judgment as a pair of relations on the two endpoints of a distinction, **same** and **diff**, subject to two conditions that any honest reading of “a judgment about a distinction” must satisfy:

- (i) **tight**: $\text{diff } a b \iff \neg \text{same } a b$ (the judgment does not abstain);
- (ii) **separating**: diff holds of the two endpoints `left`, `right` of an actual distinction (the judgment does not deny the very distinction it is about).

Theorem 6.1 (Genuine sameness is equality). *For any tight, separating judgment, using only reflexivity and symmetry of **same**, the relation **same** is forced to coincide with the decidable equality that the act-generated inductive structure already carries.*

Formalized as `genuine_judgment_same_is_equality` and `comparison_is_derived_not_primitive` in `Foundation/PrimitiveRecognitionCalculus/PRCOnePrimitive.lean`.

Two points about the hypotheses, because they are what the referee is really asking about. First, reflexivity and symmetry are not substantive content about distinction; they are constitutive of what a sameness verdict is. Reflexivity is “the act did not fire on a thing against itself.” Symmetry is “distinction does not depend on which argument you write first.” Second, and this is the key one, *transitivity is not used in the proof*. On the two-endpoint type it is a consequence, not an input. More generally, the kernel of any discriminating map is automatically

an equivalence relation, so positing “ $\sim$ is an equivalence” is positing “ $\sim$ is agreement under the discrimination,” which Theorem 6.1 derives rather than assumes. The decidable equality itself is a metatheorem of free generation (constructor injectivity and disjointness), not an axiom; in Lean it is an `inferInstance`.

So the honest status of Section 3 is: the paper posits what the corpus proves. The repair is to replace the requirement with a citation. That is editorial, and it strengthens rather than weakens the paper, because the equivalence stops being a hypothesis the reader must grant.

7 Where the forcing stops

I think the reason to trust the “forced” claims above is that the program is equally explicit about what is *not* forced. A foundation that announces a single primitive owes a sharp boundary, and over the last passes that boundary has itself become a set of theorems.

The unit of scale is a gauge. The Law of Logic forces the *form* of the recognition cost, the one-parameter family $\lambda_c(x) = \frac{1}{2}(x^c + x^{-c}) - 1$. It does not force the unit c . Every member of the family satisfies the genuine cost laws, and, sharper, every member satisfies the exact non-calibration hypotheses of the uniqueness theorem, namely reciprocity, normalization, the reciprocal cost law (the d’Alembert composition identity), and continuity on the positives. Calibration, the normalization of the leading log-curvature to 1, holds if and only if $c = 1$. So calibration is the single hypothesis that pins J , and the unit is the one irreducible gauge choice.

Formalized as `calibration_is_the_only_hypothesis_pinning_J` and `calibration_unit_not_forced_by_cost` in `PRCCalibrationIndependence.lean`; every member satisfies the cost law via `costLambda_coshAddIdentity` and `costLambda_satisfiesCompositionLaw`, and `costLambda_isCalibrated_iff` gives the $c = 1$ characterization.

The continuum is independent. Distinction forces the rational field $\mathbb{Q}_\delta$ and stops there. The completion of $\mathbb{Q}_\delta$ into $\mathbb{R}$ is a separate commitment. The order-completeness (least-upper-bound) property is not entailed by the cost laws plus the field structure: any countable cost-closed subfield of $\mathbb{R}$ fails it while $\mathbb{R}$ satisfies it, so completeness is exactly the content the continuum adds and nothing the cost forces.

Formalized as `completeness_is_exactly_the_continuum`, `completeness_not_forced_by_genuine_cost_laws`, and `countable_subfield_not_complete` in `PRCCompletenessIndependence.lean`.

Quotients are extra information. This is the same boundary that answers (Q3), restated as a limit. The forced object is the free, minimal one. Modular arithmetic $\mathbb{Z}/n$, for instance, is a quotient of the trace monoid: it declares that after n distinctions you are back where you started. That declaration is information distinction does not contain. So $\mathbb{Z}/n$ is permitted but not forced, and the thing that *is* forced is precisely the relation-free tower $\mathbb{N}_\delta \subset \mathbb{Z}_\delta \subset \mathbb{Q}_\delta$.

8 New elements formalized in the current program

The objection arrived while the program was closing exactly the gaps it names. The following are formalized, each with no project-local axioms and no `sorry`, and each audited to depend only on Lean’s standard kernel axioms `{propext, Classical.choice, Quot.sound}`.

1. **Orbit forcing (answers Q1, Q2).** The Lawvere NNO characterization of the trace orbit (`logicNat_isNNO`), uniqueness up to canonical isomorphism (`IsNaturalNumberObject.equiv`), the same property for every admissible realization (`realizationOrbit_isNNO`), and the Boolean parity-collapse theorem that separates orbit from carrier (`interpret_eq_parity`, `interpret_collapses`).

2. **Recognition-law faithfulness.** The derivation of a faithful $\mathbb{N}$ -copy from the single recognition law (`interpret_injective`, `nativeOrbitEquiv`), non-vacuous across the music, biology, narrative, and ethics realizations (`four_canonical_domains_faithful`), and under the canonical cost J (`jcostLadder_faithful`).
3. **One primitive (answers Q3).** Sameness forced to be the act’s decidable equality (`genuine_judgment_same`, `comparison_is_derived_not_primitive`).
4. **Calibration independence.** The cost form is forced and the unit is a gauge, proved against the full law set including the reciprocal cost law (`calibration_is_the_only_hypothesis_pinning_J`).
5. **Completeness independence.** The continuum is a separate commitment, sharpened to “completeness is exactly the continuum” (`completeness_is_exactly_the_continuum`).
6. **Inevitability parses.** Faithful realizations of δ inside hereditarily finite set theory, full ZFC over Mathlib’s `ZFSet` with the axiom of infinity modelled, Martin-Löf type theory, and the topos of sets, together with the distinction dichotomy “degenerate or realizes δ ” (`distinction_dichotomy`). These show the δ core appears, by the foundation’s own discrimination mechanism, in settings that never mention traces.
7. **Shared countable field.** The canonical cost and every named RS constant live in one countable, $\exp/\log/J$ -closed proper subfield of $\mathbb{R}$ (`cost_and_constants_share_one_countable_field`), so the cost dynamics never touch the uncountable reals.

9 What remains to complete

I will not claim more than is proved. The following are open, in decreasing order of how directly they bear on the referee’s question.

1. **Paper-level reconciliation (editorial, near-term).** Replace the two places where *Arithmetic from Distinction* asserts what the corpus proves: cite Theorems 3.1–3.2 in the trace section so the successor reads as the initial-object property rather than a formation rule taken on faith, and cite Theorem 6.1 in Section 3 so the equivalence requirement reads as a derived fact. This closes the self-containment gap the referee correctly identified.
2. **Minimality of the constants’ field (open).** The shared field is proved to be a countable $\exp/\log/J$ -closed field containing $\pi, e, \varphi, \alpha^{-1}$. Proving it is the *smallest* such field would upgrade “the construction can avoid the continuum” to “the construction’s minimal carrier is pinned.”
3. **The $\delta \rightarrow$ RS-chain bridge (open, central).** The orbit forcing and the cost forcing meet in the middle. Wiring them end to end, so that φ from self-similarity, the eight-tick cadence, $D = 3$, and the constants run on the countable field fed by the δ cost, is the live Universal Forcing front (`Foundation/UniversalForcing/`). This is what would make “a single primitive for physics” hold as a chain rather than as a handshake.
4. **Empirical residual, unrelated to this paper (open).** In the mass sector a neutral-closure mass ratio is predicted at φ^4 against a measured value near 5.79, an order-one residual. I will not fit it; a genuine fix needs a derived geometric ingredient, not a tuned constant.

10 Summary

The referee asked which part comes from distinction and which from the trace construction. The answer is that the trace construction is not a separate source of content; it is the initial, relation-free realization of one iterated act, and initiality is the precise meaning of “forced.” The successor is the iterate of the generator, and its Peano behavior is derived from the single recognition law that distinct distinctions are distinguishable, formalized without reference to traces and surviving even a two-element carrier. The equivalence relation on sameness is derived from tightness and separation, not assumed. And the forcing stops, provably, at the rational field: the unit of scale is a gauge and the continuum is independent. The reader’s reframing, “arithmetic from a free monoid generated by one distinction symbol,” is correct, and once *free* is read as *initial*, it is the theorem the paper should have stated outright.

Lean modules referenced. `Foundation/UniversalForcing/NaturalNumberObject.lean`, `Foundation/UniversalForcing/PrimitiveRecognitionCalculus/PRCOnePrimitive.lean`, `PRCCalibrationIndependence.lean`, `PRCCompletenessIndependence.lean`, `PRCCostOnField.lean`, and the inevitability parses `PRCFullZFCParse.lean`, `PRCTypeTheoryParse.lean`, `PRCCategoryTheoryParse.lean`, `PRCSetTheoryParse.lean`. All build clean and pass the axiom audit in `scripts/prc_delta_frontier_audit.lean`.